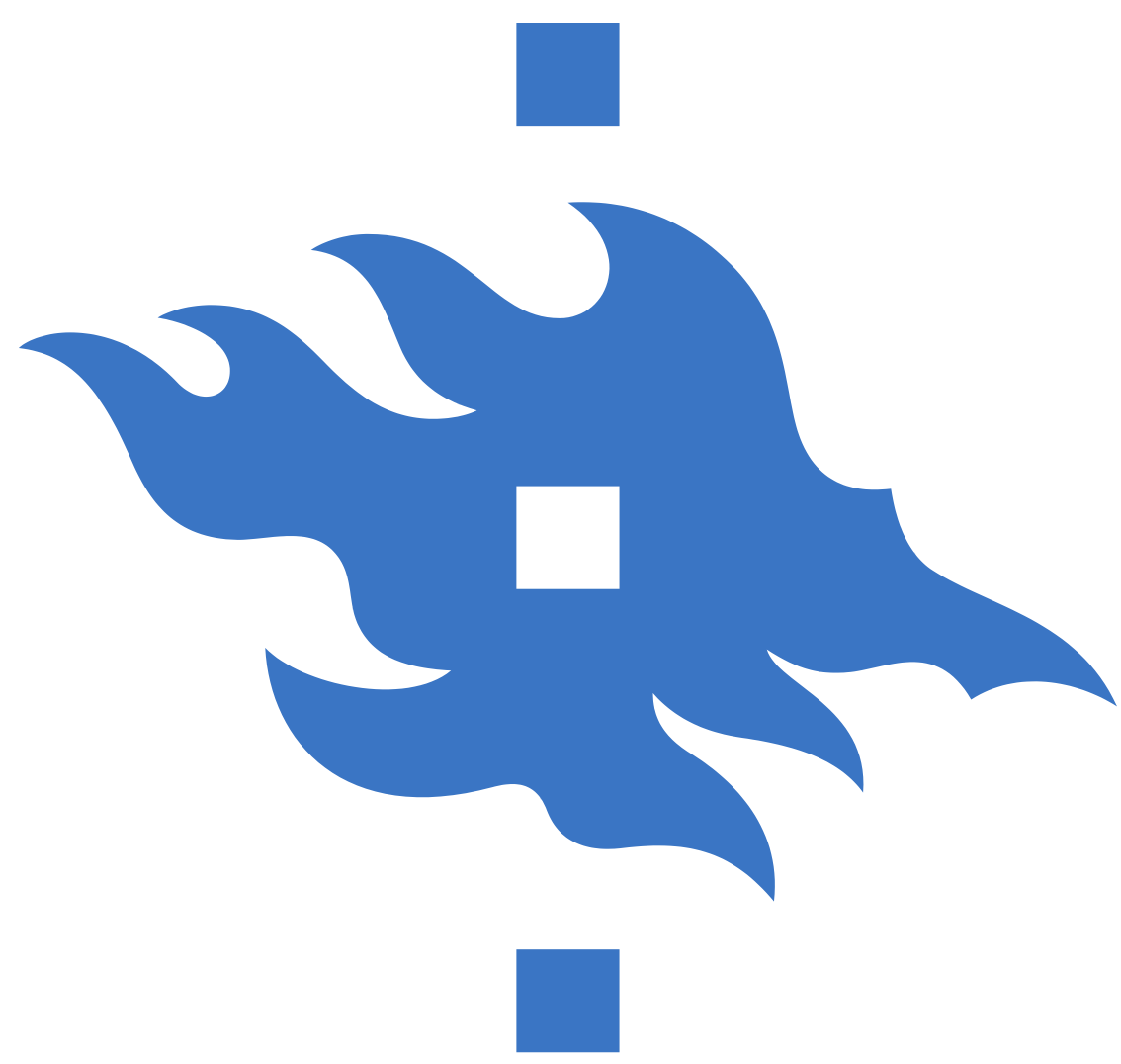


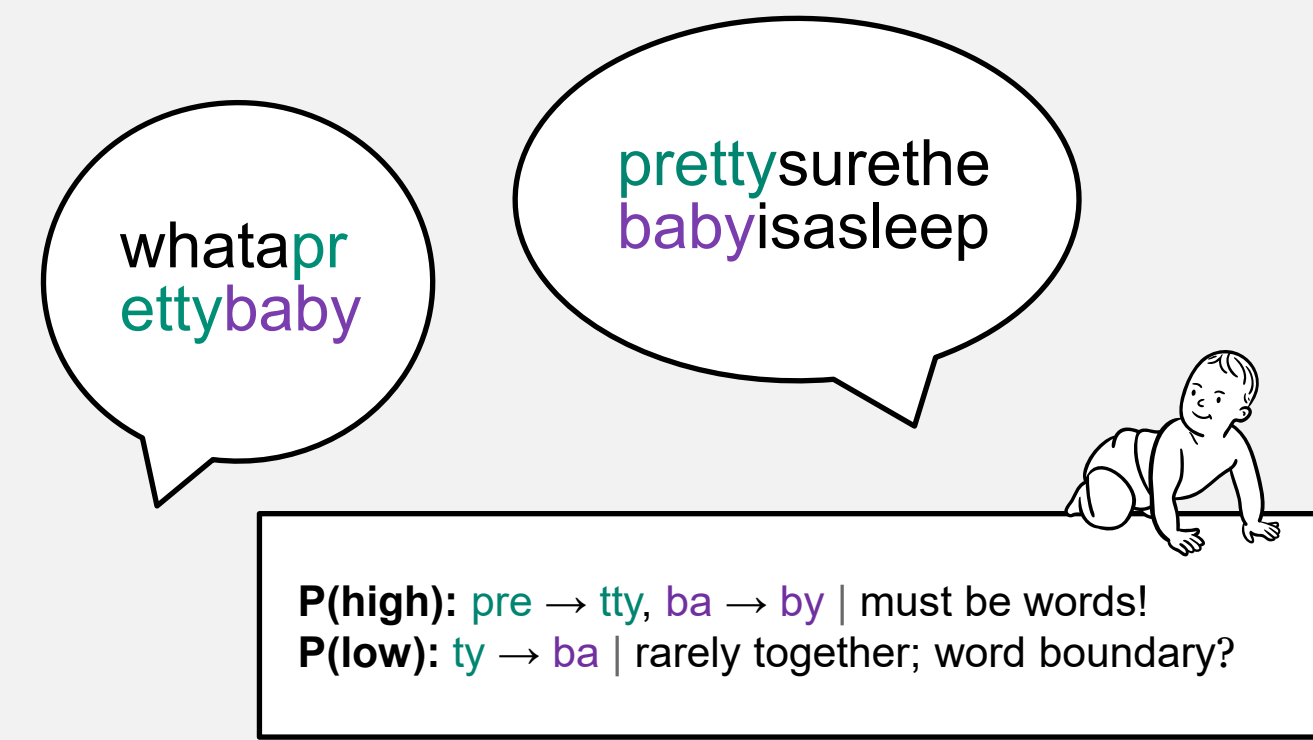


MULTILINGUAL EXPERIENCE HAS NO EFFECT ON STATISTICAL LANGUAGE LEARNING

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INTRODUCTION



- Statistical learning is assumed to be a central mechanism supporting language acquisition.
- Previous research suggests that multilinguals are at an advantage in new language learning.

Could this advantage be due to better statistical learning ability?

Previous evidence is inconclusive and has limitations: test paradigms include additional cues or may not control for previous familiarity of syllable combinations, allowing learning to extend beyond simple tracking of transitional probabilities.

CONCLUSION

- We asked whether multilingualism affects statistical language learning.
- We used an entropy-based score capturing language use and diversity (LHQ3), with additional data-driven analyses capturing AoA.
- Participants demonstrated robust statistical language learning in both auditory and visual modalities.
- However, no relationship between the MLD indices and statistical language learning scores was found.
- Our results suggest that learning novel probabilistic patterns from linguistic input is not a skill that improves with multilingual experience.

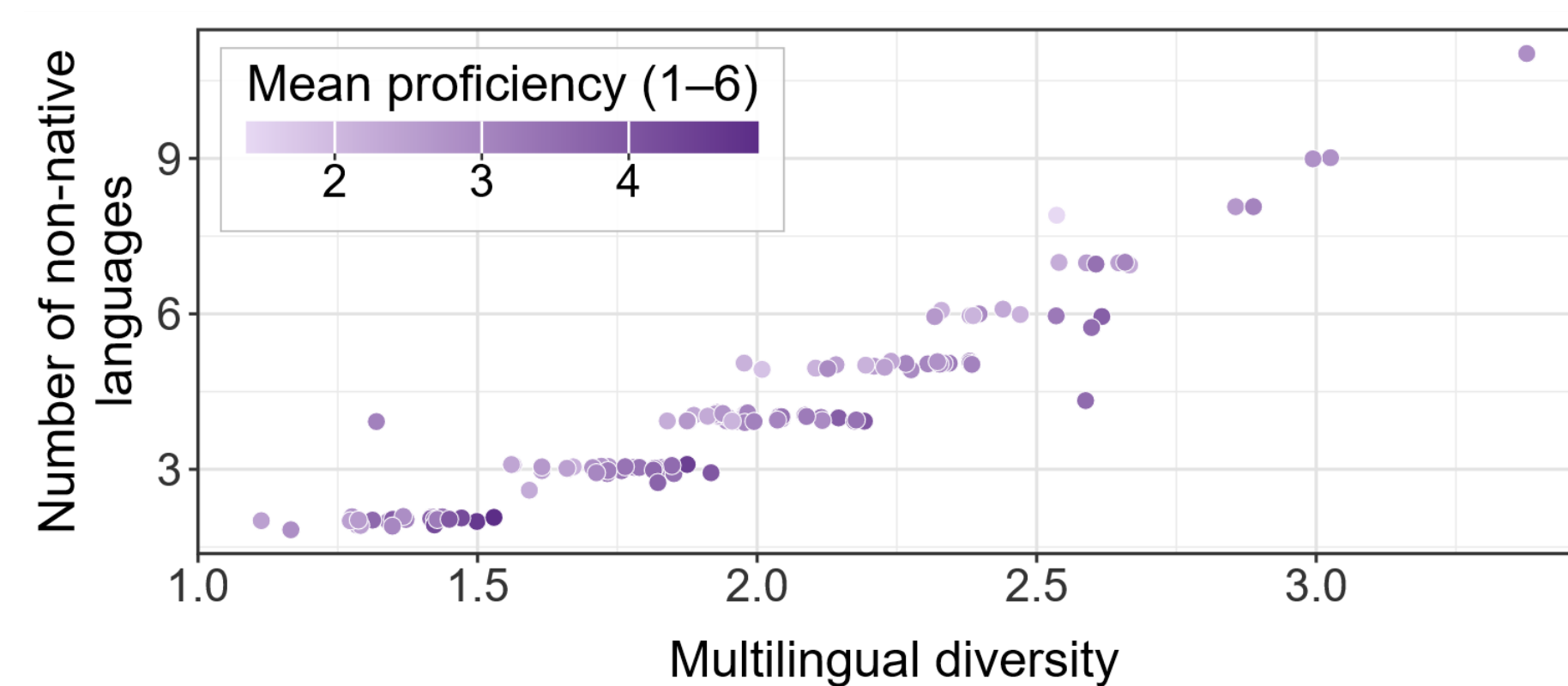
METHODS

127 Finnish L1 young adults with experience in 2–11 additional languages completed statistical language learning tasks in an online experiment.

MODELING MULTILINGUALISM

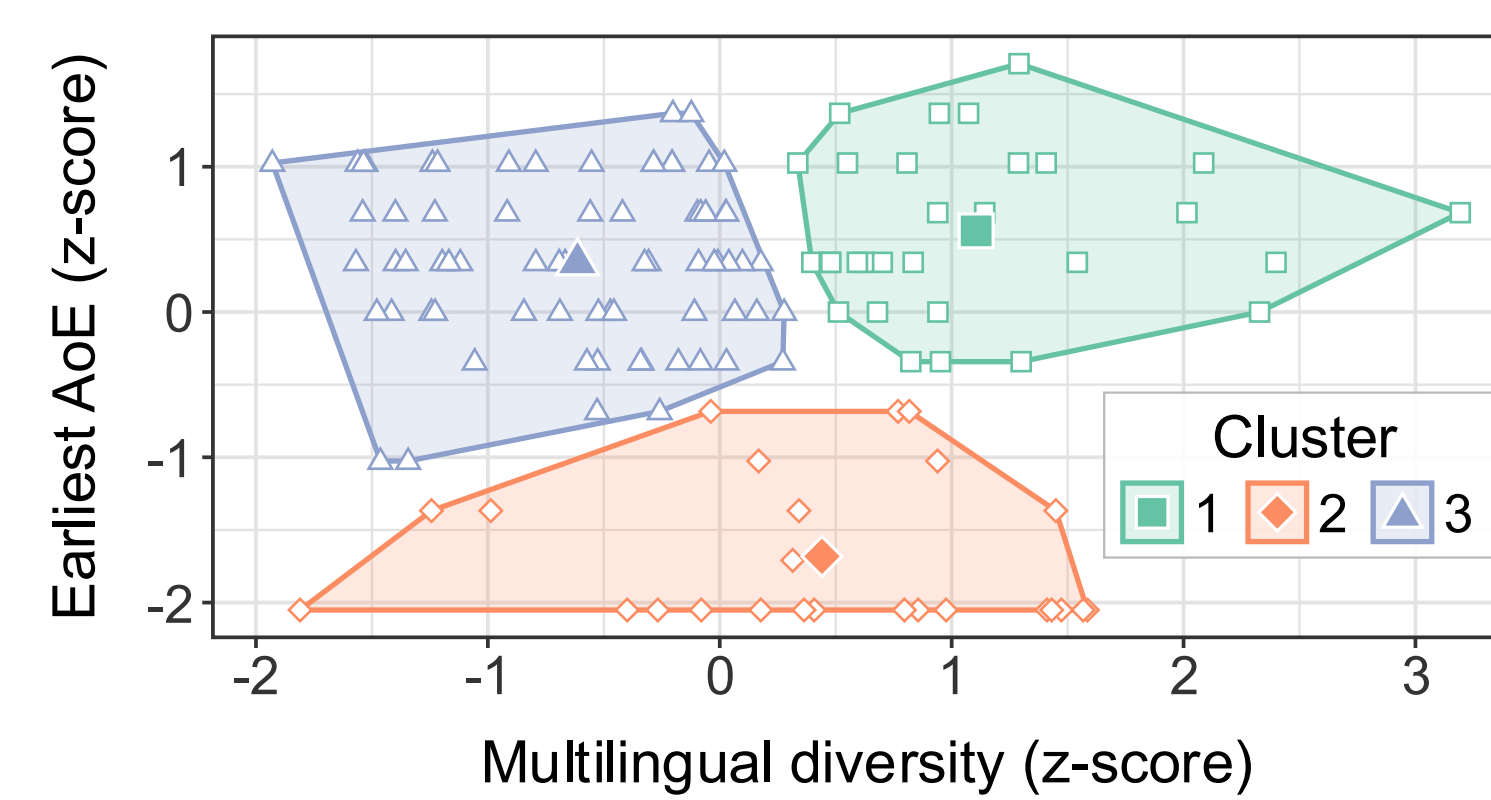
Multilingual experience was quantified using an entropy-based multilingual diversity (MLD) index derived from the Language History Questionnaire (LHQ3.0 [1]), capturing proficiency, number of known languages, and balance of language use. To incorporate earliest age of L2 exposure, we additionally ran a k-means cluster analysis.

MLD, PROFICIENCY, AND NUMBER OF L2S



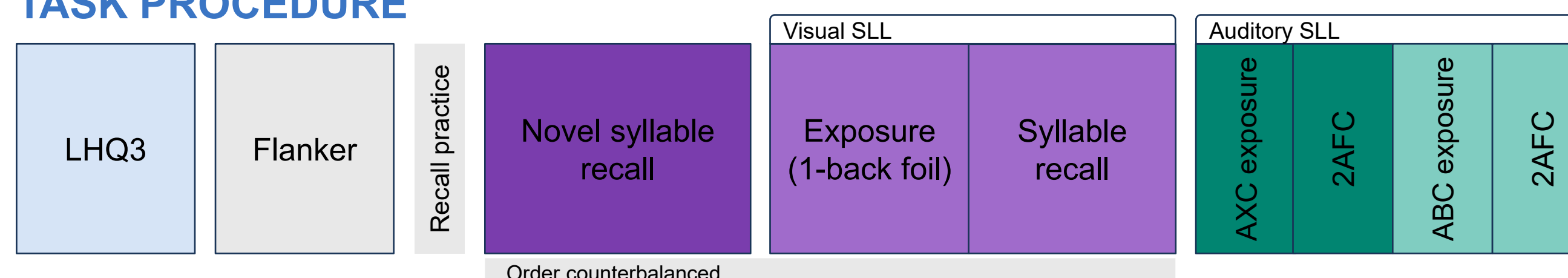
min	mean	max
• MLD 1.11 • 2 non-native languages • Mean prof. at 2.5	• MLD 1.97, SD = 0.44 • 4 non-native languages • Mean prof. at 3	• MLD 3.38 • 11 non-native languages • Mean prof. at 2.86

MLD AND AoE



	n	mean MLD	mean AoA
1 Late-High	30	2.45 [2.11–3.38]	7.63 [5–11]
2 Early-Low	25	2.16 [1.17–2.67]	1.08 [0–4]
3 Late-Low	73	1.69 [1.11–2.09]	7.03 [3–10]

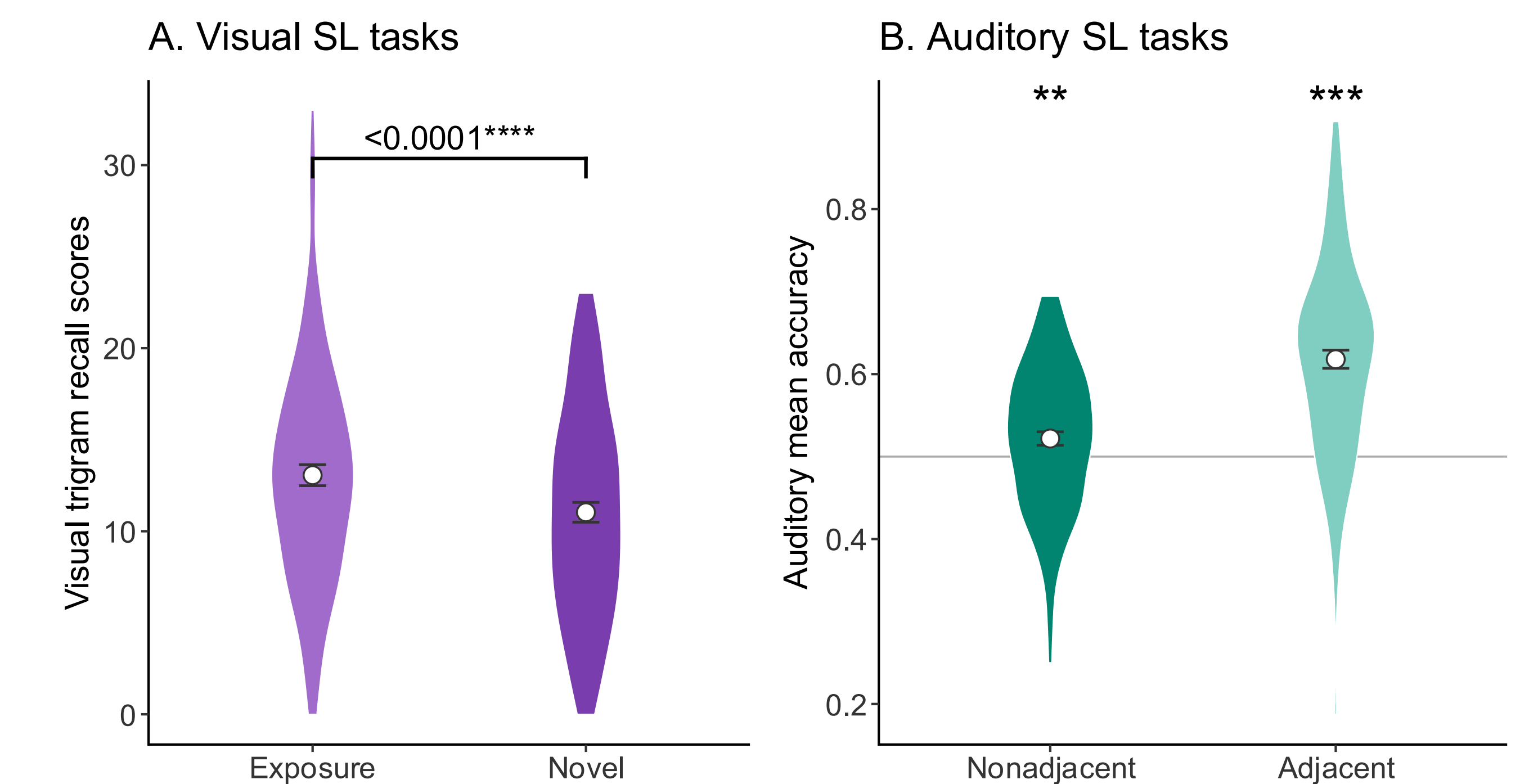
TASK PROCEDURE



In the statistical learning tasks, participants were exposed to meaningless words presented as continuous streams of syllables, separately presented in the auditory and visual modality. In the auditory task, learning was assessed with an auditory recognition task (2AFC). In the visual task, learning was assessed with a syllable recall task (SICR [2]).

RESULTS

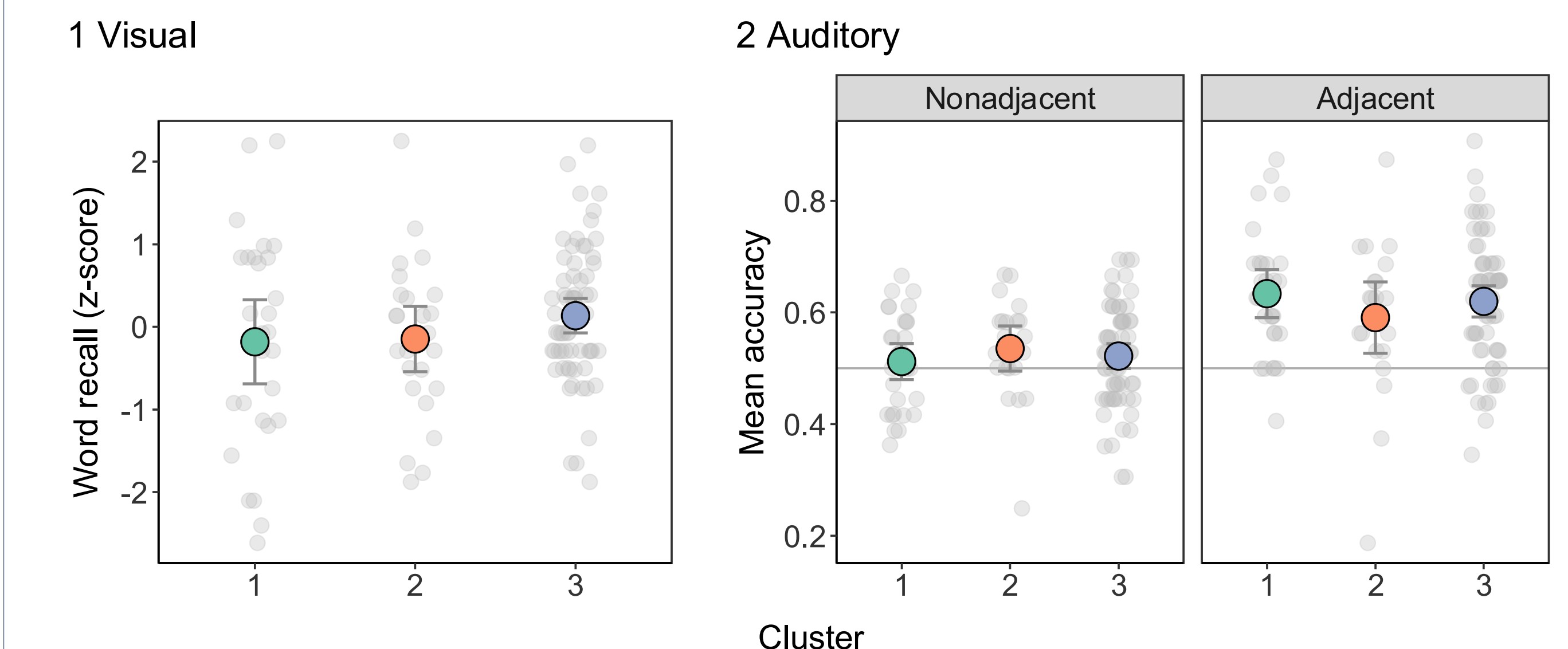
VISUAL AND AUDITORY STATISTICAL LANGUAGE LEARNING



Group-level learning. A. Visual statistical learning task (paired t-test comparing scores for exposed and novel trigrams) and B. auditory statistical learning tasks (one-sample t-tests against 50% chance level for nonadjacent and adjacent dependencies). The grey line indicates chance level. White points are grand means with error bars showing ± 1 SEM; violin plots show score distributions. **** $p < .0001$, *** $p < .001$, ** $p < .01$.

NO EFFECT OF MULTILINGUALISM ON STATISTICAL LANGUAGE LEARNING

- Correlational analyses and LMEs showed no relationship between MLD indices and learning outcomes.
- There were no differences in learning outcomes between multilingual profiles incorporating AoA.



Multilingual profiles and learning. K-means clustering on multilingual diversity and earliest age of exposure (AoE) classified participants into (1) **Late-High**, (2) **Early-Low**, and (3) **Late-Low** multilinguals. We compared learning outcomes between profiles for both tasks: 1 Visual, normalised difference scores (post-exposure – VWM); 2 Auditory, mean accuracy in the 2AFC tasks (grey line at .50). Coloured points are means with 95% CI; grey points are individual scores.